Week 9 Level 3: Advanced

How Do Machines Learn?

AI finds patterns in examples – and copies their mistakes.

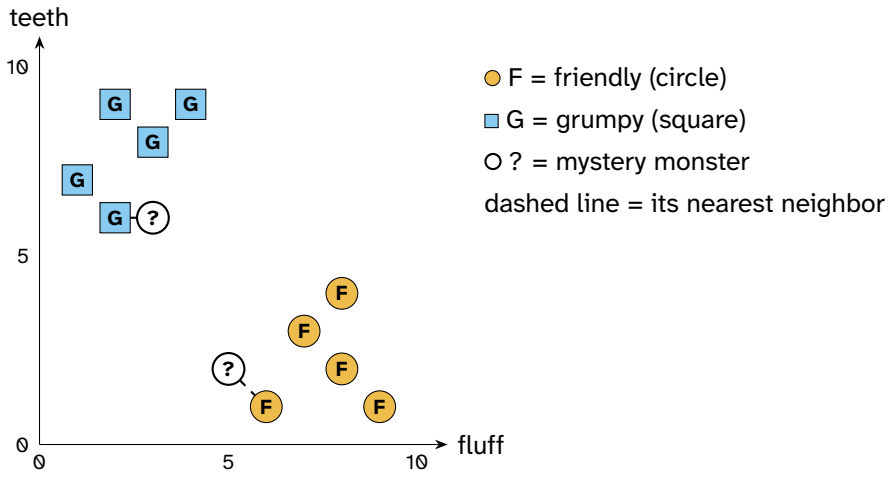
What you will learn

- The difference between a program that follows **rules** and one that **learns from examples**.
- How a **nearest-neighbor classifier** sorts things, and how to measure its **accuracy**.
- Why unbalanced or wrong examples make an AI unfair (**bias**).
- How a tiny text generator writes sentences – and why chatbots can be confidently wrong.

Rules or examples?

A program with **rules** does exactly what you typed: `if teeth > 5: print("grumpy")`. A **learning** program gets lots of **examples** with the right answers (**labels**) and finds the pattern itself. The things it measures are called **features**.

Our monsters have two features: **fluff** (0–10) and **teeth** (0–10). Put every example on a chart. To guess a **mystery** monster, find the closest example and copy its label. That is a **nearest-neighbor classifier**.



The mystery at (5, 2) is closest to a friendly monster, so the guess is *friendly*. Distance is measured with Pythagoras: `math.sqrt(dx * dx + dy * dy)`.

Is it any good? Train, then test

Split the examples into two piles. The program learns from the **training** pile. The **test** pile stays secret until the end; then we count how many test cards it labels correctly.

$$\text{accuracy} = \frac{\text{correct guesses}}{\text{number of test cards}}$$

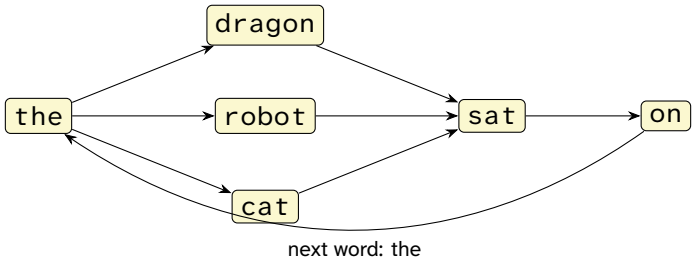
Testing on cards it learned from would be like seeing the exam answers first.

Bug hunt: Bad examples, bad predictions

If the training pile has 8 grumpy monsters and only 1 friendly one, the program calls most friendly monsters grumpy. The code is fine – the *data* is the bug. This is called **bias**. When real AI decides things about people (faces, voices, job applications), biased examples can treat whole groups unfairly. That is why people must check the data, not just the code.

A tiny text generator

A **Markov chain** learns just one thing from a text: *after each word, which words came next?*



To write, it starts with a word and keeps picking a *random* next word from the list. The sentences *sound* like the training text, but the machine does not understand them at all. Today's chatbots are enormously bigger and cleverer, but they also predict likely next words – which is why they can sound sure and still be wrong.

The five big ideas of AI

Big idea	What it means	Today
1 Perception	Computers sense the world with cameras, microphones, sensors.	a camera measuring fluff?
2 Representation & reasoning	The world is stored as data the computer can reason about.	each monster = (fluff, teeth)
3 Learning	Computers find patterns in examples.	nearest-neighbor classifier
4 Natural interaction	Talking and writing like people is very hard.	Markov story machine
5 Societal impact	AI can help or harm people.	bias, privacy, deepfakes

Exercise 9.1: Monster sorter

Folder: `week09/ex-monster-sorter`. Finish `distance(a, b)` and `nearest(training, mystery)`. Run it and look at the chart. What does it guess for (5, 5)? Do you agree?

Exercise 9.2: Fair or biased?

Folder: `week09/ex-fair-or-biased`. Finish `label_counts()` and `accuracy()`. Fair training accuracy: _____ Biased training accuracy: _____ Why is the biased one worse?

Exercise 9.3: Markov story machine

Folder: `week09/ex-markov-story`. Finish `build_chain(words)`: each word maps to the list of words that came right after it (keep repeats!). Copy your silliest generated sentence:

Exercise 9.4: My project pitch (Milestone 1)

For the next three weeks you will *Build Something Big*. Read the Project brief, then fill in your pitch. Finish it at home.

Project title: _____

Option (circle one): grid game generative art app data/AI app

In one sentence, it is...: _____

Two classes I will need: _____

Inheritance or composition: _____

Data structure (dict / stack / queue / grid): _____

Named algorithm: _____

File it will save: _____

The smallest fun version: _____

Digital citizenship & safety: Using AI wisely and safely

AI can be wrong, even when it sounds sure – check important facts with a trusted source or adult. **Never share personal information** with a chatbot: no real name, address, school, passwords, photos or secrets; what you type may be stored. **Deepfakes** are fake pictures, voices or videos made by AI: ask "Who made this, and why?" before you believe or share, and never make fakes of real people. **Bias**: AI is only as fair as its examples.

Vocabulary box

feature Something we measure about an example, like fluff or teeth.

label The right answer for an example, like "friendly".

training data The labeled examples a program learns from.

test data Examples kept secret during training, used to check the guesses.

accuracy Correct guesses divided by the number of test examples.

classifier A program that sorts things into groups (labels).

bias When unbalanced or unfair data makes a program treat some cases unfairly.

deepfake A fake photo, voice or video made with AI.

Reflection

Where have you met AI this week (videos, games, phones, homework)?

One thing you will never type into a chatbot, and why:

Week 9 Level 3: Advanced

How Do Machines Learn?

15–20 minutes. Do both Core parts; Challenge and Creative are optional.

Name (nickname is fine): _____ Date: _____

★ Core (everyone): Part A: Fruit detective (about 10 minutes)

Open `3-advanced/exercises/week09/hw-fruit-detective/starter.py`. One card in `TRAINING` has the wrong label: a lemon was labeled "apple". With **k = 1** the classifier copies the single closest card. With **k = 3** the three closest cards **vote**.

1. Finish `k_nearest(training, point, k)`: sort the cards by distance (`sorted(..., key=...)` from week 5) and keep the first k.
2. Finish `vote(labels)` with a counting dictionary.
3. Run it and `check_my_work.py`. k = 1 says: _____ k = 3 says: _____

★ Core (everyone): Part B: Finish your project pitch (Milestone 1, about 5 minutes)

Finish the pitch card in workbook Exercise 9.4. Check it against the Project brief: two or more classes (one with inheritance or composition), a dictionary/stack/queue/grid, a named algorithm, a saved file. Bring it to the next session.

★★ Challenge (want more?): How big should k be?

In the fruit detective, try k = 1, 3, 5 and 13 for the mystery fruit (5, 4) and for (2, 8). What happens when *every* card votes (k = 13)? Why is a very big k a bad idea too?

Creative (make it yours): Spot the AI

This week, notice three places where AI might be at work (video suggestions, photo filters, voice assistants, game opponents…). For one of them, write: what are its *examples* (training data)? Could it be unfair to someone? Do not type anything personal into any AI tool to find out!

Self-check: tick each box when it is true

- ☐ I can explain the difference between rules and learning from examples.
- ☐ I know why we test on cards the program did not learn from.
- ☐ I can explain how bad or unbalanced examples lead to unfair predictions.
- ☐ I never share personal information with a chatbot.
- ☐ My project pitch card is complete.